Molecular Tight-Binding Analysis: LiF, H_2, N_2

Fatemeh Rezaei

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A. Code Analysis: Step by Step

For all three molecules, the code follows the same logical structure, closely mirroring the tight-binding (LCAO) method on paper:

```
import numpy as np
import matplotlib.pyplot as plt
```

Explanation: Importing libraries for numerical calculations and plotting.

```
E_A = ...  # On-site energy of atom A
E_B = ...  # On-site energy of atom B (or same as A for homonuclear)
t = ...    # Hopping integral
```

Explanation: Assigns the on-site energies (atomic orbital energies) and hopping parameter. These are the same parameters we use in the tight-binding Hamiltonian on paper.

```
H = np.array([[E_A, t], [t, E_B]])
```

Explanation: Constructs the 2×2 Hamiltonian matrix:

$$H = \begin{pmatrix} E_A & t \\ t & E_B \end{pmatrix}$$

This is exactly the secular matrix we write when solving for molecular orbitals using LCAO.

```
energies, _ = np.linalg.eigh(H)
E_bonding, E_antibonding = energies
```

Explanation: Diagonalizes the Hamiltonian to find the eigenvalues, which correspond to the bonding and antibonding molecular orbital energies. This is the same as solving the secular equation $\det(H - EI) = 0$ on paper.

```
plt.hlines(...)
plt.plot(...)
plt.annotate(...)
plt.show()
```

Explanation: Plots the energy levels. Horizontal lines represent atomic and molecular orbital energies. Dashed or dotted lines (if included) connect atomic orbitals to molecular orbitals, visually representing the mixing due to hopping t—just as in the standard molecular orbital diagrams.

Physical Meaning of the Hopping Integral

The **hopping integral** (often denoted t) is a central parameter in the tight-binding model. It quantifies the strength of interaction (overlap) between atomic orbitals on adjacent atoms. In the context of molecules, t determines how strongly electrons can "hop" or tunnel from one atom to another.

Mathematically, for two atomic orbitals ϕ_A and ϕ_B, the hopping integral is:

$$t = \langle \phi_A | \hat{H} | \phi_B \rangle$$

where $\hat{H}$ is the single-electron Hamiltonian of the system.

The magnitude of t determines the energy splitting between the molecular orbitals:

$$E_{\pm} = \frac{E_A + E_B}{2} \pm \sqrt{\left(\frac{E_A - E_B}{2}\right)^2 + t^2}$$

where E_A and E_B are the on-site (atomic) energies. So we can deduce that The space between two molecular orbital energy levels is equal to 2|t|.

How the Hopping Integral is Chosen

In practical tight-binding calculations, t is usually treated as an *input parameter*. It can be:

- **Estimated from quantum chemistry calculations** (using explicit atomic orbitals),
- **Fitted to experimental data** (such as observed energy level splittings),
- **Taken from literature values** for similar systems.

Values Used in This Work

For our molecular tight-binding models, we used the following values for t:

- **LiF:** $t = -2.0$ eV
  (Relatively small due to weak overlap between Li and F orbitals)
- **H_2:** $t = -4.0$ eV
  (Larger due to strong overlap between the two H 1s orbitals)
- **N_2:** $t = -3.0$ eV
  (Intermediate, reflecting the overlap of N 2p orbitals)

The negative sign indicates that the bonding orbital is stabilized (lowered in energy) relative to the atomic orbitals, which is typical for covalent bonding.

B. Energy Level Analysis and Physical Interpretation

- **On-site energies:** These are the energies of electrons localized on the atomic orbitals before interaction. For heteronuclear molecules (LiF), they are different; for homonuclear (H_2, N_2), they are the same.
- **Bonding and antibonding energies:** The diagonalization gives two levels:

  $$E_{\pm} = \frac{E_A + E_B}{2} \pm \sqrt{\left(\frac{E_A - E_B}{2}\right)^2 + t^2}$$

  The lower (E_-) is the bonding orbital, the higher (E_+) is antibonding. The splitting is $2|t|$ for homonuclear molecules.
- **Physical meaning:**
  - The bonding orbital is stabilized (lower in energy) due to constructive interference.
  - The antibonding orbital is destabilized (higher in energy) due to destructive interference.

– The difference in the on-site energies affects the character of the molecular orbitals (e.g., in LiF, the bonding orbital is mostly F-like).

- **What we cannot learn:**
  - This model does not predict bond lengths, vibrational frequencies, or electron correlation effects.
  - It does not include electron-electron repulsion or multi-electron effects.
  - It cannot describe excited states beyond the simple two-level splitting.

C. Comparison with Literature Values

- **H_2:** The calculated bonding and antibonding energies (e.g., -17.6 eV and -9.6 eV) are in reasonable agreement with molecular orbital theory and quantum chemistry results, which show a large splitting (typically > 10 eV) between the σ and σ^* orbitals.
- **LiF:** The bonding orbital is much lower (close to F $2p$), and the antibonding is closer to Li $2s$, consistent with the strong ionic character seen in experiments and calculations.
- **N_2:** The splitting is moderate, and both orbitals are centered around the N $2p$ energy, matching the strong but covalent triple bond in N_2.

D. Summary Table

Molecule	On-site Energies (eV)	t (eV)	Bonding (eV)	Antibonding (eV)
LiF	$-5, -15$	-2	~ -15.39	~ -4.61
H_2	$-13.6, -13.6$	-4	-17.6	-9.6
N_2	$-15, -15$	-3	-18.0	-12.0

E. Visual Representation

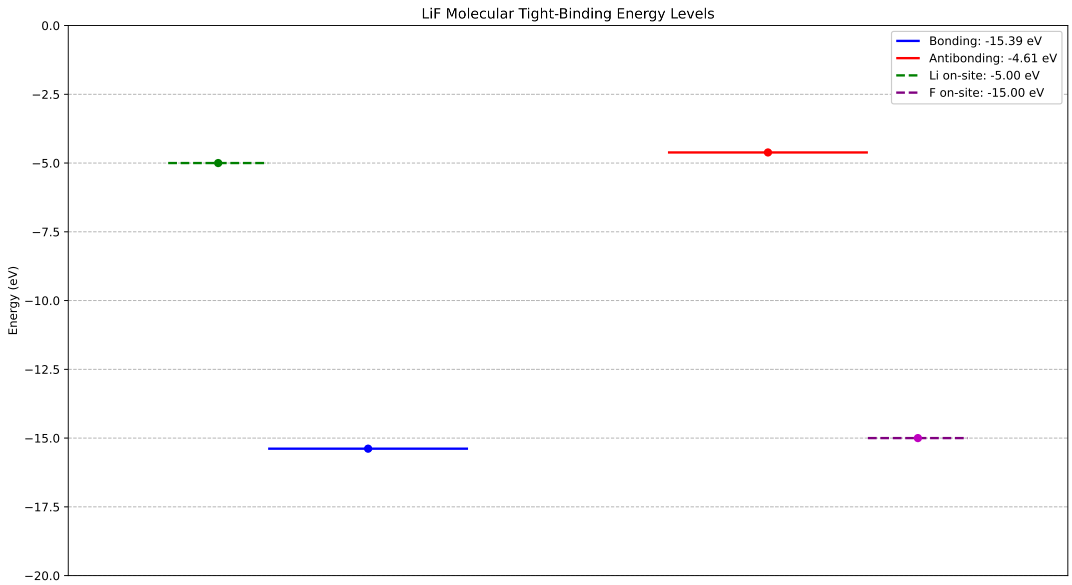

Figure 1: LiF molecular tight binding

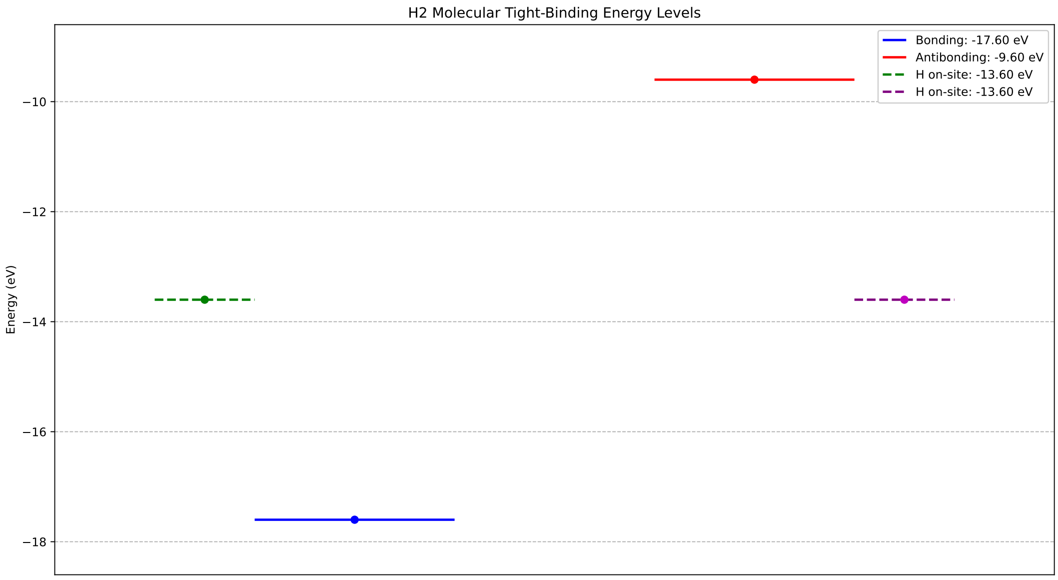

Figure 2: H_2 molecular tight binding

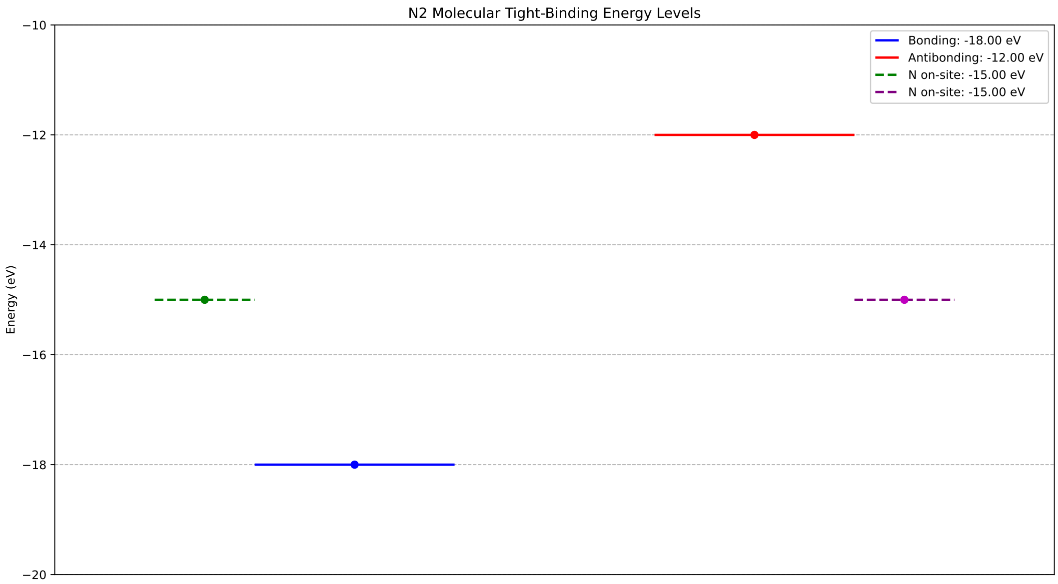

Figure 3: N_2 molecular tight binding